Embedded Systems -- Guided Notes ANSWER KEY

Unit tp-2.13 | CompTIA Network+ N10-009 Obj. FC0-U71 2.13 | N10-008 FC0-U71 2.13

For instructor use / distribute after students complete the notes.

Question 1

An embedded system is a _______ computing system built into a device for a specific _______ function. Unlike a general-purpose computer, an embedded system is not designed to run _______ different applications. Examples include _______, _______, and _______ (name three devices).

Answer: Dedicated; purpose-built; many; microwave ovens, traffic lights, washing machines (any three).

Question 2

A microcontroller is the primary chip inside many embedded systems. It integrates a _______ (processor), _______ (program storage), _______ (working memory), and I/O interfaces all into a single _______. This makes it compact, cheap, and power-efficient for dedicated tasks.

Answer: CPU; flash memory (ROM); RAM; chip (IC).

Question 3

An RTOS (Real-Time Operating System) is designed to process inputs and produce outputs within a _______ deadline. Hard real-time systems must meet every deadline or the system _______. Examples include engine control units in cars and medical _______ where a missed deadline could cause harm.

Answer: Guaranteed (strict); fails (malfunctions); devices (pacemakers).

Question 4

Firmware is software stored in _______ (non-volatile memory, often flash or ROM) inside a hardware device. It controls the basic operation of the device at a _______ level. Unlike standard software, firmware _______ (can/cannot) usually be updated without special tools or vendor authorization, and updates carry _______ risk if interrupted.

Answer: Non-volatile memory (flash/ROM); low (hardware level); can; high (bricking the device).

Question 5

Embedded systems are common in _______ (vehicles with computerized fuel and emissions control), _______ (insulin pumps, defibrillators), _______ (assembly robots, PLCs), and _______ (routers, switches). In critical systems, embedded software is subject to rigorous _______ testing before deployment.

Answer: Automotive; medical devices; industrial (manufacturing); networking equipment; safety/reliability.

Question 6

The three main constraints of embedded systems that limit them compared to general-purpose computers are limited _______ (CPU speed), limited _______ (RAM and flash), and limited _______ (battery or power supply). These constraints require developers to write highly _______ code with minimal overhead.

Answer: Processing (compute); memory; power; optimized (efficient).

Question 7

An SoC (System on a Chip) integrates all major components of a computer -- _______, _______, and _______ controllers -- into a single silicon die. SoCs are common in _______ phones, _______ devices, and embedded systems because they reduce size, cost, and power consumption.

Answer: CPU, GPU, memory; I/O; smartphones; IoT.

Question 8 -- Real World Application

REAL WORLD: A car manufacturer recalls 50,000 vehicles because of a critical software bug in the embedded engine control unit (ECU) that can cause unintended acceleration. Unlike a PC software bug, the fix cannot simply be emailed to users. Explain what challenges the manufacturer faces in updating the embedded firmware and describe two methods they could use to deliver the fix.

Answer: Challenges: (1) ECUs run firmware stored in non-volatile flash memory, not on a hard drive -- updating requires specialized hardware interfaces (OBD-II port, J-Tag) or a specific programming protocol. (2) A failed or interrupted firmware flash can brick the ECU (leave the car inoperable), so the update must be safe, checksummed, and reliable. (3) Not all owners bring their car to a dealer promptly. Method 1 -- Dealer OBD-II flash: customers bring the vehicle to a dealer where a technician connects a programmer to the OBD-II diagnostic port and flashes the updated firmware file -- safe, controlled environment. Method 2 -- OTA (Over-the-Air) update via cellular: if the vehicle has a telematics module with a data connection (like Tesla), the manufacturer pushes the signed firmware update wirelessly. The ECU validates the signature before applying the update, and it applies only when the vehicle is parked with the engine off to avoid interruption.